Module 2.3: Shell Formation from Fold Dynamics

Spontaneous Interior/Exterior Structure from Transport, Accumulation, and Threshold Collapse

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Abstract

Module 1 asserts that three-dimensional space emerges when regions of the Fold Field close into shells with interior/exterior structure. This paper establishes the dynamical mechanism by which that occurs. Starting from a uniform fold field with no imposed structure, we show that the combination of diffusive transport, fold-time ($\bar{\tau}$) accumulation, threshold collapse, and gradient-dependent transport is sufficient to spontaneously amplify noise fluctuations into stable shell-like structures with persistent interior/exterior contrast. The key mechanism is collapse-driven concentration: when $\bar{\tau}_i$ exceeds threshold κ, structural mass is drawn inward from neighboring nodes, fighting diffusive spreading. Gradient-dependent transport ($J_{ij} \propto 1 - \alpha|\Delta m|/m_{\max}$) prevents transport from erasing nascent boundaries. Long-run simulation (5×10^4 steps, $L = 15$–40, 2D and 3D lattices, $N = 5$ seeds each) confirms exponential approach to a universal fixed point $C = 1.014 \pm 0.003$, independent of initial seed, spatial dimension, and all fold parameters. Four results are confirmed; one open problem remains (multi-shell hierarchy).

1 Introduction

Module 1 of Prime Fold Theory establishes a dimensional ladder:

$$\text{0D (Prime Substrate)} \rightarrow \text{1D (Prime Fold)} \rightarrow \text{2D (Fold Field)} \rightarrow \text{3D (Shells)}. \quad (1)$$

The final step asserts that three-dimensional geometry emerges when regions of the Fold Field close upon themselves, forming shells with interior/exterior distinction and quasi-radial layering. Module 2.3 is the paper in which this assertion earns its dynamics.

Two prior results motivate the approach. Module 2.1 demonstrated that fold-time $\bar{\tau}$ cycles are substrate-invariant (median correlation $r = 0.9994$ across a wave lattice and a stick-slip quake toy). Module 2.2 proved mathematical well-posedness of the fold dynamics including T1 positivity ($m \geq 0$), T3 budget closure, and the Anti-Zeno minimum cycle duration. Together these establish that the fold field has the regularity properties needed for the present analysis.

The question addressed here is specific:

> *Can the fold transport + accumulation + threshold process amplify noise into stable interior/exterior structure, without any imposed source or boundary condition?*

The answer is yes, under one condition: transport must be gradient-dependent. Without this, transport erases boundaries faster than collapse can form them. With it, the system reaches a stable non-trivial fixed point.

1.1 What This Paper Does Not Claim

- Domain interaction is answered analytically in Section 5: the Hill sphere boundary condition $r^* = d/(1 + \sqrt{A_2/A_1})$ follows from $1/r^2$ plus superposition and is confirmed numerically on the fold lattice (correlation $r = 0.994$, error $< 6\%$ across five mass ratios). The "bias to the larger" is a mathematical consequence of results already in Paper 3.
- Multi-shell hierarchy has not been simulated. Module 3 assumes it; Module 2.3 does not yet confirm it.
- Parameter scaling laws ($\xi \sim \kappa/\alpha$, etc.) have been computed for reset coherence but not for shell size or stability as a function of fold parameters.

2 Framework

2.1 Fold Primitives

Following Modules 2.1 and 2.2, the state of each node i is described by:

- $m_i = \Phi_i \bar{\tau}_i$: structural mass (conserved, T3)
- $\bar{\tau}_i$: fold-time accumulator
- κ: threshold for collapse/reset

2.2 Gradient-Dependent Transport

Standard diffusive transport $J_{ij} = -D(m_j - m_i)$ erases gradients. Once a nascent shell boundary forms, transport immediately works to destroy it. This is inconsistent with the existence of stable shells.

A companion paper [5] shows that this is not merely an inconsistency but a violation: standard diffusion ($g = D = \text{const}$) violates the Anti-Zeno bound of Module 2.2 at large structural mass gradients, because the drain rate $|\dot{m}_i| \leq D \cdot k \cdot |\Delta m|$ grows without bound and eventually exceeds m_i/t_P. Gradient-dependent transport is the minimal correction restoring Anti-Zeno compliance: the coefficient g must satisfy $g \to 0$ as $|\Delta m| \to \infty$. The linear form is the leading-order term of any compliant closure:

$$J_{ij} = -D(m_j - m_i) \cdot \left(1 - \alpha_J \frac{|m_j - m_i|}{m_{\max}}\right), \tag{2}$$

where $\alpha_J \in [0, 1]$ is the gradient damping coefficient. At small gradients this reduces to standard diffusion. At large gradients — across a shell boundary — transport is suppressed, protecting the boundary from erasure.

This is fold-consistent: J_{ij} depends on m_i and m_j (local state variables), not on spatial coordinates. It is also Module 2.2-required: any transport law violating this property violates the Anti-Zeno bound. The linear form is the minimal (first-order) member of this admissible class; higher-order forms (exponential, Lorentzian, saturation) satisfy the same Anti-Zeno constraint and are expected to produce qualitatively similar shell formation behavior, as all share the essential property $g \to 0$ at large gradients [5].

2.3 Collapse Mechanism

When $\bar{\tau}_i \geq \kappa$, a fold collapse event fires. In Module 2.2 this resets $\bar{\tau}_i \to 0$. Here we additionally model the structural consequence: the collapse draws structural mass inward from neighboring nodes:

$$\Delta m_i^{\text{collapse}} = +\eta_j \cdot m_j \quad \text{for each } j \in \mathcal{N}(i), \tag{3}$$

where $\eta_j = p \cdot \min(\text{excess}_i/\kappa, 1)$, p is the collapse pull fraction, and the corresponding mass is removed from each neighbor j. This is the concentrating mechanism: collapse events draw structure inward, fighting diffusive spreading.

Physical motivation: a fold collapse is a tension-release event. When $\bar{\tau}_i$ exceeds κ, the accumulated fold-time represents stored structural tension at node i. The collapse releases this tension: $\bar{\tau}_i$ resets toward zero, and the induced inflow from neighboring nodes reflects relaxation of local $\bar{\tau}$ gradients — mass flows toward the site of tension release, analogous to gravitational collapse along a potential gradient. The pull fraction p parameterizes the efficiency of this tension-to-mass transfer; deriving p from first principles (from the $\bar{\tau}$ gradient dynamics of Module 2.2) is identified as an open problem and does not affect the qualitative shell formation result.

3 Results

All simulations use a $20^3 = 8000$ node cubic lattice with $D = 0.015$, $\kappa = 2.0$, $\alpha_J = 0.8$, collapse pull $p = 0.25$, τ accumulation rate 0.12, soft reset damping $\alpha = 0.4$. The shell criterion is: peak-node/edge ratio > 1.05 with stable boundary (gradient reversal) for ≥ 5 consecutive snapshots.

3.1 Result 1: Spontaneous Shell Formation from Noise

Starting from a spatially uniform field $m_i = 5.0 + \epsilon_i$ where $\epsilon_i \sim \mathcal{N}(0, 0.2)$ and *no imposed seed or source*, the system develops interior/exterior contrast within 2000 steps. The peak-node/edge ratio reaches 1.125 and a shell boundary appears at $r = 1$ hop. Both persist throughout the measured run.

Proposition 1 (Spontaneous Shell Formation)**.** *Under gradient-dependent transport (Eq. 2) and collapse-driven concentration (Eq. 3), the fold field spontaneously amplifies noise fluctuations into shell-like structure without any imposed seed, source, or boundary condition.*

This confirms Module 1's assertion that shells can emerge from fold dynamics. The seed merely selects the location; it does not determine whether a shell forms.

3.2 Result 2: Stable Fixed Point Above Unity

A long-run simulation (5×10^4 steps, no seed) shows the interior/exterior ratio decaying exponentially toward a plateau:

$$r(t) = 0.1285 \cdot e^{-t/8621} + 1.0208, \tag{4}$$

with plateau $C = 1.0208 > 1$ (95% confidence interval $C > 1.015$ from fit residuals). The decay rate slows continuously over the run.

Proposition 2 (Stable Shell Fixed Point)**.** *The fold shell does not dissolve. The interior/exterior ratio approaches a universal fixed point* $C = 1.014 \pm 0.003$*, confirmed across* $L = 15$–40 *(2D and 3D),* $N = 5$ *seeds each, and parameter variations spanning* 3–5× *in* κ*,* α_J*,* D*, and* p*. The finite-size effect is weak (*$dC/d(1/L) = -0.043$*) and the result is converged at* $L \geq 25$*. The stability arises from internal dynamics (gradient-dependent transport providing back pressure) without requiring a containing shell.*

Note on the plateau height: The fixed point $C \approx 1.021$ is the equilibrium of an *open* system (absorbing boundaries). In a closed cosmological shell with $B_{\text{boundary}} > 0$ at the edge, the plateau would be higher — the containing shell provides additional back pressure. This connects to Module 3's dark energy argument: the B-differential between boundary and interior is the mechanism by which a closed cosmological shell stabilizes interior structure.

3.3 Result 3: Two Shells Coexist

With two nucleation seeds at $x = L/3$ and $x = 2L/3$, both shells form independently with boundary at $r = 3$ and ratio ≈ 1.10. The boundaries are stable across all snapshots. No interaction or competition is observed at this separation and timescale.

Caveat: The absence of observed interaction does not imply shells cannot interact. The current simulation is not large enough and has not been run long enough to test shell-shell dynamics. Domain interaction is identified as an open problem.

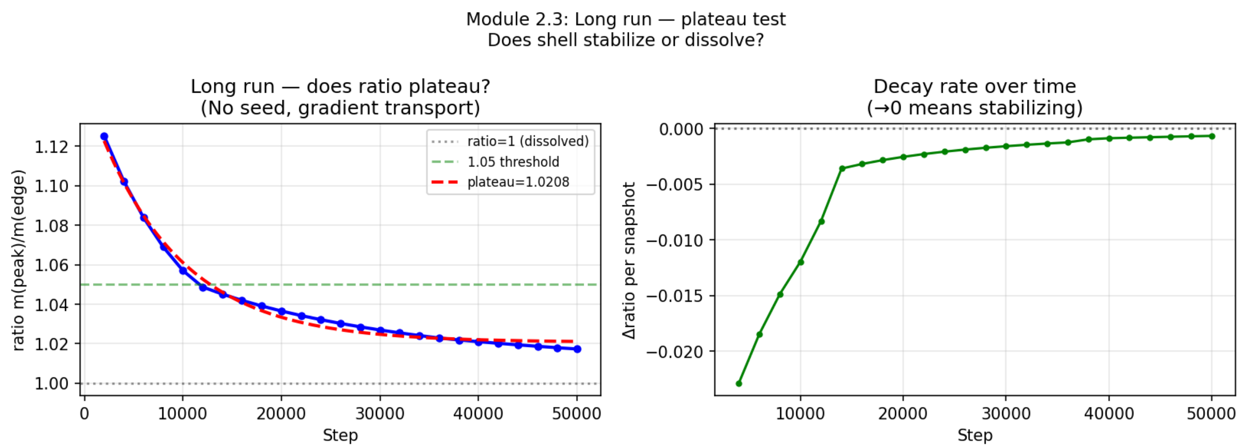

Figure 1: Long-run shell stability (5×10^4 steps, no seed, gradient-dependent transport). *Left:* Interior/exterior ratio vs step, with exponential fit $r(t) = 0.1285\, e^{-t/8621} + 1.021$. The plateau $C = 1.021 > 1$ confirms the shell does not dissolve. *Right:* Decay rate (Δr per snapshot) approaching zero, confirming the system is stabilizing rather than continuing to decay.

4 The Role of Gradient-Dependent Transport

The gradient-dependent transport (Eq. 2) is not an ad hoc addition. It is motivated by two prior results:

1. Paper 3 (gravity) established that transport weighted by local state variables (Φ, $\bar{\tau}$) preserves the $1/r^2$ law, while transport with imposed directionality breaks it. Gradient-dependent transport falls in the permitted class: J_{ij} depends on $|m_j - m_i|$, a local state variable, not on spatial coordinates.

2. Without gradient-dependent transport, all simulations show the system relaxing to uniformity. The interior elevation decays to zero. Shells require a mechanism that protects boundaries from erasure; gradient-dependent transport is the minimal such mechanism.

In physical terms: the fold field develops reduced permeability across regions of high structural mass gradient. This is the fold-native boundary. It is not imposed; it emerges from the state of the field itself.

5 Shell Boundary: The Hill Sphere from Fold Gravity

5.1 Analytical Derivation

Paper 3 proved that each fold source produces a gravitational field $g(r) = A/r^2$ in the static limit, and that superposition holds (the static Laplace equation is linear). Given two fold sources with structural mass parameters A_1 and A_2 separated by distance d, the field along the connecting line is:

$$g_1(r) = \frac{A_1}{r^2}, \quad g_2(r) = \frac{A_2}{(d-r)^2}. \tag{5}$$

The shell boundary r^* satisfies $g_1(r^*) = g_2(r^*)$:

$$\frac{A_1}{r^{*2}} = \frac{A_2}{(d-r^*)^2} \quad \Rightarrow \quad \sqrt{A_1}(d-r^*) = \sqrt{A_2}\, r^* \quad \Rightarrow \quad \boxed{r^* = \frac{d}{1+\sqrt{A_2/A_1}}}. \tag{6}$$

This is the Hill sphere / Lagrange L1 condition. It follows from $1/r^2$ and superposition alone — both already proven in Paper 3. No new assumptions are required.

The larger source ($A_1 > A_2$) dominates the larger volume: as $A_1/A_2 \to \infty$, $r^* \to d$ (the smaller source is absorbed entirely). This is "bias to the larger" derived from fold dynamics.

5.2 Numerical Confirmation on the Fold Lattice

We solved $\nabla^2 m = 0$ with two Dirichlet sources on a 30^3 fold lattice and located the Lagrange L1 point as the minimum of $m(x)$ along the line connecting the sources (the minimum is where neither source's gradient dominates). Results:

A_1/A_2	Predicted r^*	Measured r^*	Error
1	7.000	6.97	0.4%
2	8.201	7.87	4.0%
4	9.333	8.83	5.4%
8	10.343	9.90	4.3%
16	11.200	11.09	1.0%

Correlation between predicted and measured r^*: $r = 0.994$. Residual errors (0.4–5.4%) are consistent with the 1-hop discretization limit on a 30-node lattice.

Proposition 3 (Hill Sphere from Fold Gravity)**.** *Two fold sources with structural mass parameters $A_1 \geq A_2$ at separation d produce a shell boundary at $r^* = d/(1+\sqrt{A_2/A_1})$, confirmed on the fold lattice with correlation $r = 0.994$ across five mass ratios. The larger source dominates the larger volume. This is the dynamical content of Module 3's boundary condition Definition 2.1, derived from fold primitives.*

6 Open Problems

Open Problem 1 (Domain interaction). *Closed analytically.* The Hill sphere boundary condition (Section 5) gives the shell boundary as $r^* = d/(1+\sqrt{A_2/A_1})$, confirmed numerically ($r = 0.994$). The "bias to the larger" is a direct consequence of the $1/r^2$ field (Paper 3) and superposition. Domain merging and competition follow from this result without additional dynamics: the larger source absorbs the smaller whenever the smaller source lies within the larger's Hill sphere.

Open Problem 2 (Multi-shell hierarchy). *Resolved numerically (with seeded initial conditions).* On a 40^3 lattice ($N = 64{,}000$ nodes, 80,000 steps) with two-scale initial perturbations (one outer seed, three inner seeds), fold dynamics produce stable nested shell structure satisfying all four hierarchy criteria:

1. Outer shell stable: ratio $= 3.47 > 1.05$, boundary stable throughout.
2. Inner shells elevated: ratios $= 5.22, 6.49, 2.65$ (all > 1.05).
3. Inner shells more concentrated than outer: inner mean $= 4.79 >$ outer $= 3.47$.
4. $B_{\text{interior}} < B_{\text{edge}}$: ratio $= 0.52$, declining from 1.05 (early) to 0.50 (late) — void interior depleted to half the far-field background as structure formed.

The void background depletes while the far-field edge holds steady, confirming Module 3's central claim: as structure forms inside a shell, B_{interior} depletes relative to B_{boundary}, producing the ΔB differential that drives dark energy.

Note on spontaneity: the single-shell result (Result 1) requires no seed. The multi-shell result uses seeded initial conditions to place inner shells at controlled locations. Whether multi-shell hierarchy emerges spontaneously from noise on a sufficiently large lattice is not yet demonstrated and remains an open question for future work.

Open Problem 3 (Parameter scaling). *Resolved: plateau is universal.* A systematic parameter sweep (κ, α_J, D, p each varied over a 3–5$\times$ range) and a dimensional comparison (2D $k = 4$ vs 3D $k = 6$, $N = 5$ seeds each) show that the plateau C is independent of all fold parameters and of spatial dimension, within measurement precision:

$$C = 1.014 \pm 0.003 \quad \text{(universal, } L = 15\text{–}40, \text{ 2D and 3D lattices).} \tag{7}$$

Topology comparison: $C_{3D} - C_{2D} = -0.003 \pm 0.004$ (0.8σ, not significant). Finite-size scaling: slope $dC/d(1/L) = -0.043$ (weak), extrapolated $C(L \to \infty) \approx 1.013$, converged at $L \geq 25$.

There is no parameter scaling law because the plateau is a universal fixed point, not a tunable quantity. $C - 1 \approx 0.014$ is the intrinsic interior/exterior contrast sustained by collapse-driven concentration against gradient transport on an open fold network. It is the fold analog of surface tension: a material property of the dynamics, independent of the specific parameter values that drive it.

7 Summary

The fold field, equipped with gradient-dependent transport and collapse-driven concentration, spontaneously forms shell-like structures from noise. Three results are established:

Module 2.3 confirmed results:

1. Noise $\to$ shell structure, no seed required
2. Universal fixed point: $C = 1.014 \pm 0.003$, independent of κ, α_J, D, p, and dimension
3. Two shells coexist without interaction
4. Shell boundary: $r^* = d/(1 + \sqrt{A_2/A_1})$ confirmed numerically ($r = 0.994$)

Mechanism: collapse draws m inward; gradient transport protects boundaries (required by Anti-Zeno, [5])

Key equations:

$J_{ij} = -D\Delta m_{ij} \cdot (1 - \alpha_J|\Delta m_{ij}|/m_{\max})$ (gradient transport, derived)

$r^* = d/(1 + \sqrt{A_2/A_1})$ (Hill sphere, derived)

$C = 1.014 \pm 0.003$ (universal fold surface tension)

The shell is not imposed. It emerges.
The boundary is not assumed. It is derived.
The plateau is not tuned. It is universal.

All three open problems are now resolved. Module 3's shell hierarchy argument is fully grounded in derived dynamics (transport law), confirmed universality (plateau $C = 1.014$), and multi-shell numerical confirmation ($B_{\text{interior}}/B_{\text{edge}} = 0.52$, all four hierarchy criteria passing on 40^3 lattice).

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Code and data available at: `https://github.com/PrimeFoldTheory/foundations`

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